

Binomial Probability Exam Revision Guide

Prepared for Eli

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This guide covers the key binomial probability concepts and the exact practice problems we solved, complete with step-by-step translations and formulas to help you ace your exam.

The Core Toolkit: How to Translate and Solve

Before diving into problems, remember the standard binomial formula:

$$P(X = k) = \binom{n}{k} p^k q^{n-k}$$

Where:

- n : Total number of independent trials/items.
- p : Probability of success on a single trial.
- q : Probability of failure on a single trial ($1 - p$).
- k : The specific target number of successes.

English-to-Math Decoder Ring

- “At least k ” ($\geq k$): Sum from k up to n , or use the complement rule: $1 - P(X < k)$.
 - “At most k ” ($\leq k$): Sum from 0 up to k ($P(0) + P(1) + \dots + P(k)$).
 - “Fewer than k ” ($< k$): Strictly less than k (e.g., fewer than 3 means ≤ 2).
 - “Precisely / Exactly k ” ($= k$): Find that single exact probability term.
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Problem 1: The Insurance Salesman

Question: An insurance salesman sells policies to five men, all of identical age and good health. According to actuarial tables, the probability that a man of this particular age will be alive 30 years hence will be $2/3$. Find the probability that in 30 years: (i) all five men, (ii) at least one man, (iii) at least 3 men, (iv) at most 3 men, will be alive.

Variables: $n = 5$, $p = 2/3$, $q = 1/3$.

1. (i) **All five men** ($k = 5$):

$$P(X = 5) = \binom{5}{5} \left(\frac{2}{3}\right)^5 \left(\frac{1}{3}\right)^0 = 1 \times \frac{32}{243} \times 1 = \frac{\mathbf{32}}{\mathbf{243}} (\approx 0.1317)$$

2. (ii) **At least one man** ($X \geq 1$): Use the complement rule: $1 - P(X = 0)$

$$P(X = 0) = \binom{5}{0}(1)\left(\frac{1}{3}\right)^5 = \frac{1}{243} \implies 1 - \frac{1}{243} = \frac{242}{243} (\approx 0.9959)$$

3. (iii) **At least 3 men** ($X \geq 3$):

$$P(X \geq 3) = P(3) + P(4) + P(5) = \frac{80}{243} + \frac{80}{243} + \frac{32}{243} = \frac{64}{81} (\approx 0.7901)$$

4. (iv) **At most 3 men** ($X \leq 3$): Using the complement of 4 or 5: $1 - [P(4) + P(5)] = 1 - \frac{112}{243} = \frac{131}{243} (\approx 0.5391)$

Problem 2: The Multiple-Choice Quiz

Question: A student writes a five question multiple-choice quiz. Each question has four possible responses. The student guesses at random for each question. Calculate the probability for each possible score from 0 to 5.

Variables: $n = 5$, $p = 1/4 = 0.25$, $q = 3/4 = 0.75$. Formula: $P(k) = \binom{5}{k}(0.25)^k(0.75)^{5-k}$

- **Score = 0:** $\binom{5}{0}(1)(0.75)^5 = \frac{243}{1024} (\approx 0.2373)$
- **Score = 1:** $\binom{5}{1}(0.25)(0.75)^4 = \frac{405}{1024} (\approx 0.3955)$
- **Score = 2:** $\binom{5}{2}(0.25)^2(0.75)^3 = \frac{135}{512} (\approx 0.2637)$
- **Score = 3:** $\binom{5}{3}(0.25)^3(0.75)^2 = \frac{45}{512} (\approx 0.0879)$
- **Score = 4:** $\binom{5}{4}(0.25)^4(0.75)^1 = \frac{15}{1024} (\approx 0.0146)$
- **Score = 5:** $\binom{5}{5}(0.25)^5(1) = \frac{1}{1024} (\approx 0.0010)$

Problem 3: Short-Sighted Students

Question: Thirty percent of students at a large university are known to be short-sighted. If twenty students are picked at random, find the probability that at most two of them are short-sighted.

Variables: $n = 20$, $p = 0.3$, $q = 0.7$. Target: $P(X \leq 2) = P(0) + P(1) + P(2)$.

$$P(0) = \binom{20}{0}(0.3)^0(0.7)^{20} \approx 0.00080$$

$$P(1) = \binom{20}{1}(0.3)^1(0.7)^{19} \approx 0.00684$$

$$P(2) = \binom{20}{2}(0.3)^2(0.7)^{18} \approx 0.02785$$

$$P(X \leq 2) = 0.00080 + 0.00684 + 0.02785 = \mathbf{0.03548} (\approx 3.55\%)$$

Problem 4: Committee Attendance

Question: There are 10 members on a committee. The probability of any member attending a randomly chosen meeting is 0.9. The committee cannot do business if more than 3 members are absent. What is the probability that 7 or more members will be present on a given date?

Variables: $n = 10$, $p = 0.9$, $q = 0.1$. Target: $P(X \geq 7) = P(7) + P(8) + P(9) + P(10)$.

$$P(7) = \binom{10}{7} (0.9)^7 (0.1)^3 \approx 0.05740$$

$$P(8) = \binom{10}{8} (0.9)^8 (0.1)^2 \approx 0.19371$$

$$P(9) = \binom{10}{9} (0.9)^9 (0.1)^1 \approx 0.38742$$

$$P(10) = \binom{10}{10} (0.9)^{10} (0.1)^0 \approx 0.34868$$

$$P(X \geq 7) = \mathbf{0.9872} \text{ (exact fraction: } \frac{617,003,001}{625,000,000} \text{)}$$

Problem 5: Unbiased Coin Tosses

Question: An unbiased coin is tossed 10 times. Find the probability that there are fewer than 3 heads tossed.

Variables: $n = 10$, $p = 0.5$, $q = 0.5$. Target: "Fewer than 3" means $X \leq 2$ ($P(0) + P(1) + P(2)$).

$$P(X < 3) = \frac{\binom{10}{0} + \binom{10}{1} + \binom{10}{2}}{1024} = \frac{1 + 10 + 45}{1024} = \frac{56}{1024} = \frac{\mathbf{7}}{\mathbf{128}} (\approx 5.47\%)$$

Problem 6: ACME Faulty Widgets

Question: ACME Inc. produces widgets sold in boxes of 8. 2

Variables: $n = 8$, $p = 0.02$, $q = 0.98$. Target: "At least one" $\implies 1 - P(0)$.

$$P(X \geq 1) = 1 - \binom{8}{0} (0.02)^0 (0.98)^8 = 1 - (0.98)^8 \approx \mathbf{0.1492} (\approx 14.92\%)$$

Problem 7: Brian's Pizza Choices

Question: Brian eats pizza every night (7 days). Each night he picks mushroom with probability 0.35. Find the probability that he eats between 2 and 4 (inclusive) mushroom pizzas this week.

Variables: $n = 7$, $p = 0.35$, $q = 0.65$. Target: $P(2 \leq X \leq 4) = P(2) + P(3) + P(4)$.

$$P(2) = \binom{7}{2} (0.35)^2 (0.65)^5 \approx 0.29848$$

$$P(3) = \binom{7}{3} (0.35)^3 (0.65)^4 \approx 0.26787$$

$$P(4) = \binom{7}{4} (0.35)^4 (0.65)^3 \approx 0.14424$$

$$P(2 \leq X \leq 4) = 0.29848 + 0.26787 + 0.14424 = \mathbf{0.71059} (\approx 71.06\%)$$

Problem 8: Covington High School C1 Module

Question: At Covington High School, 20

Variables: $n = 15$, $p = 0.2$, $q = 0.8$. Target: $P(X = 4)$.

$$P(X = 4) = \binom{15}{4}(0.2)^4(0.8)^{11} = 1,365 \times 0.0016 \times 0.085899 \approx \mathbf{0.1876} \text{ } (\approx 18.76\%)$$